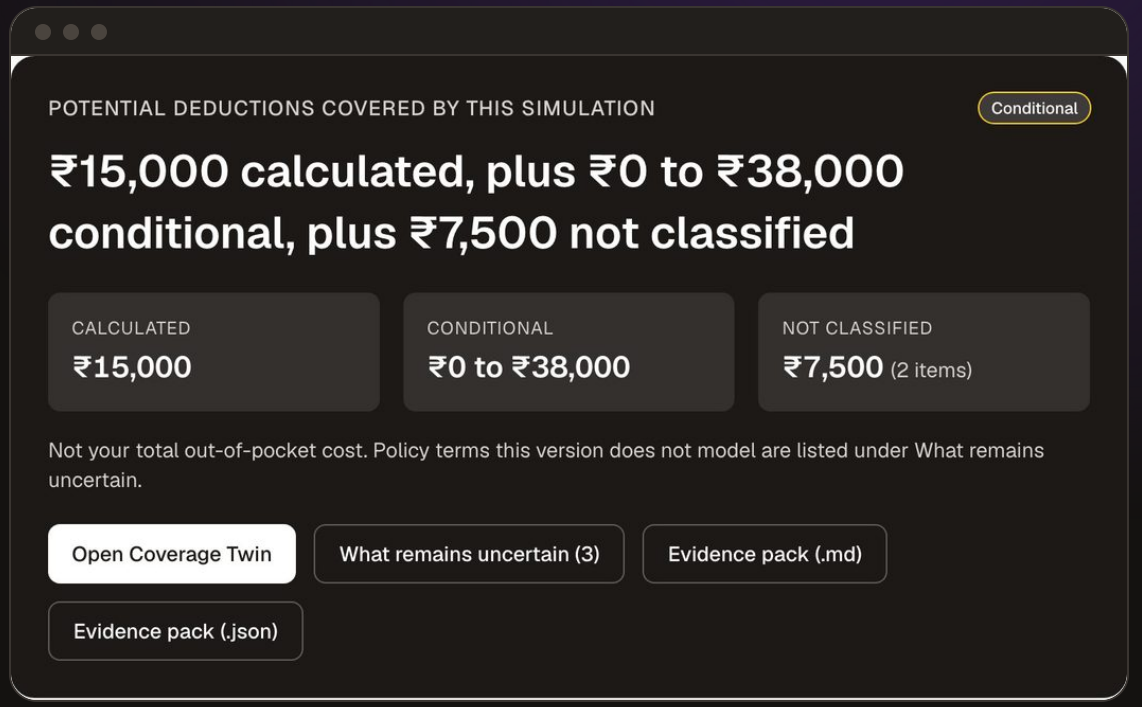
TRACK 2 · AI-POWERED FINANCIAL JOURNEYS

Claim Sach

Coverage Twin: know what your policy might leave you paying, *before* admission.

Claim Sach turns health-insurance policy wording into an interactive hospital-bill simulation, showing which potential deductions are calculated, which remain conditional, and exactly why.

Pushkar Mondal · Founding Engineer, Xeliport

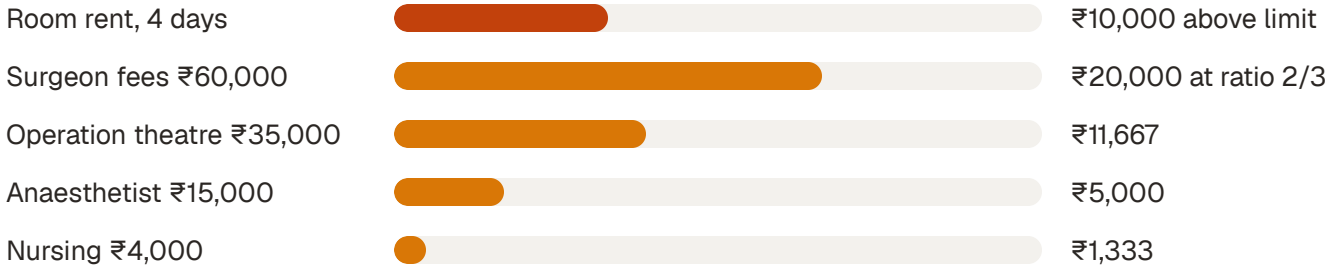


The first thing a user sees: calculated, conditional and unclassified amounts kept apart, never folded into one total.

The policy's cost shows up at the hospital

Room-rent limits, proportionate deductions, ICU limits, co-payments, sub-limits and exclusions surface at admission, not at purchase. Pick a room above the eligible rate and many policies pay the surgeon, the anaesthetist, nursing and the operation theatre only in the proportion **eligible ÷ actual rate**. It is not about the room.

One room choice, four items that are not the room



Worked example on a synthetic policy and estimate, computed by the prototype and checked by hand in docs/deck/CALCULATION-AUDIT.md (section "Synthetic wording"). Not a market statistic.

₹2,500 a day

the room-rent gap: ₹7,500 chosen against ₹5,000 eligible

₹48,000

potential deductions from that one choice; ₹38,000 of it on items that are not the room

4 of 11 lines

affected under the policy's definition; 3 excluded by it; 2 the wording never classifies

Which items are touched depends on the policy's own definition, on facts only the hospital knows, and on clauses the policyholder has never read. General explainers stop at the concept. The gap we are testing is applying *your* wording to *your* bill.

Documents in, itemised deductions out, every step on screen

1 **Policy documents**
Wording and schedule, synthetic or real, labelled which.

2 **Structured rules**
Each rule cites its clause and quotes it in full.

3 **Scenario simulation**
Room, rate, days, the hospital's own line items.

4 **Itemised potential deductions**
Room excess, ICU excess, proportion, co-payment.

5 **Source clause**
Page-cited, verbatim, beside every number.

6 **Explicit uncertainty**
What is open, what is at stake, who can settle it.

0:00
Overview
₹15,000 calculated, ₹0 to ₹38,000 conditional, ₹7,500 not classified. Not a total, and it says so.

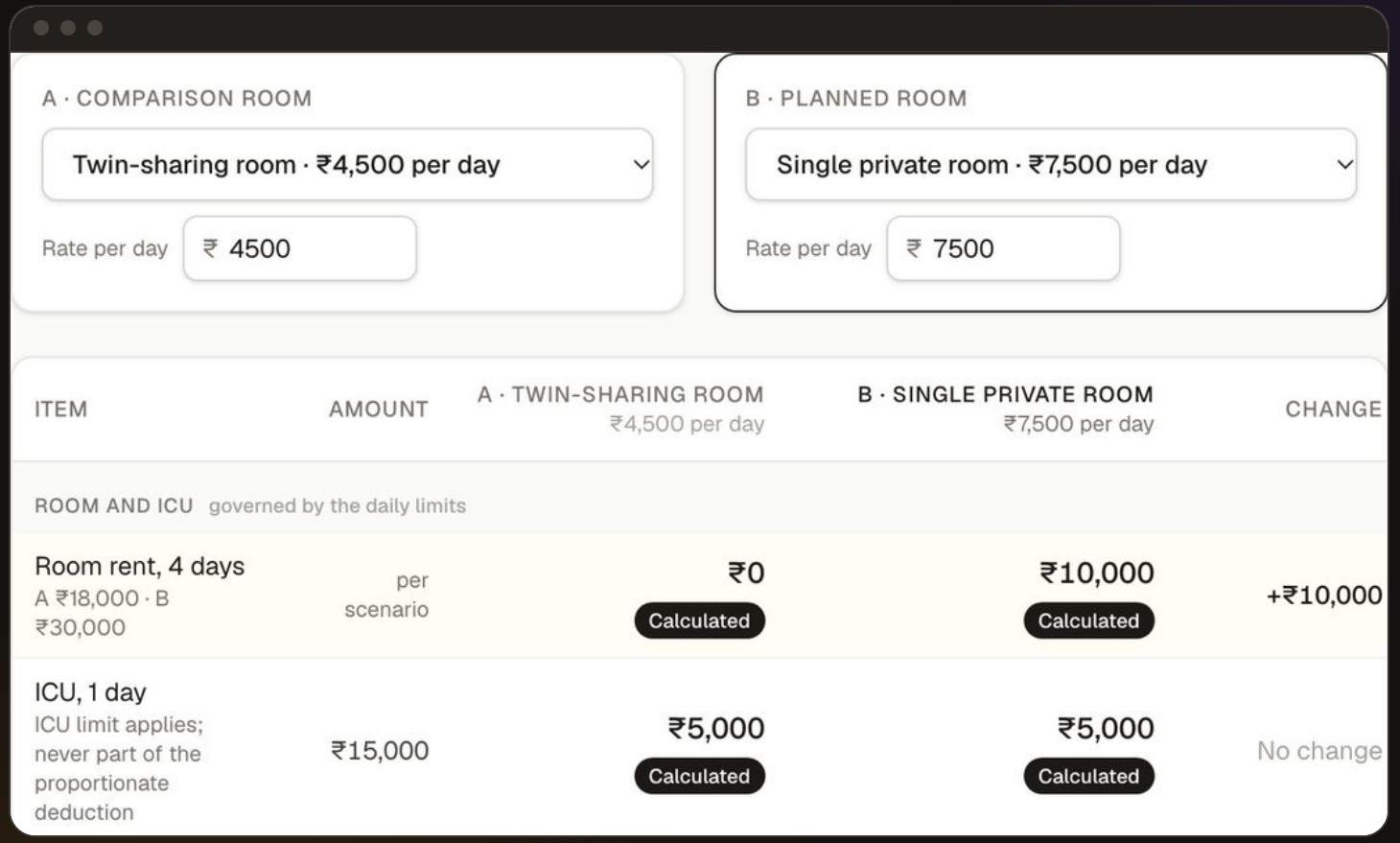
0:25
Twin
Change the room; the itemised breakdown updates at once with the cause rule per row.

1:35
Uncertain
Ask the hospital, ask the insurer, resolve now; plus what this version does not model at all.

3:00
Real wording
Switch to Arogya Sanjeevani and watch the same engine refuse to guess where the wording is silent.

Not a chatbot. Not a payout guarantee. A scenario estimate of the deductions it covers, with its unknowns on screen. Six screens, two wordings, one command, offline. Everything on these slides runs today; what is proposed is marked as such on slide 11.

Change your room. See the difference.



Twin-sharing (A) against single private (B), same policy, same estimate. The room row changed: ₹0 against ₹10,000 above the limit, clause C1. The ICU row did not: its limit never depends on the room.

1. **One policy, one estimate, two rooms**
   Only the rows a rule touched light up. The rest stay exactly as they were.

2. **Every amount opens its evidence**
   Clause and page, inputs with where they came from, the formula, the result.

3. **Ranges, not guesses**
   Until the hospital confirms differential billing, the surgeon's deduction is ₹0 to ₹20,000.

4. **Unknown is never zero**
   Lines the wording never classifies stay out of every total, and the panel says who can settle them.

Every number opens its evidence

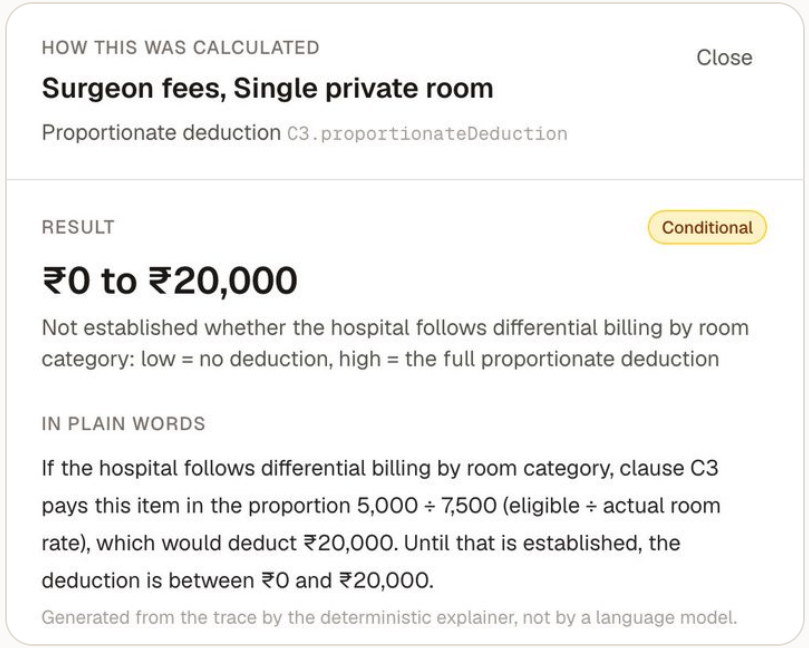
What a trace holds

- The rule, by id and name.
- Each input with its unit and where it came from: the documents, your upload, or you.
- The source clause, quoted verbatim with document, version and page.
- The formula as text, and by id in the tests.
- The result in paise, with its state and any open assumption by code.

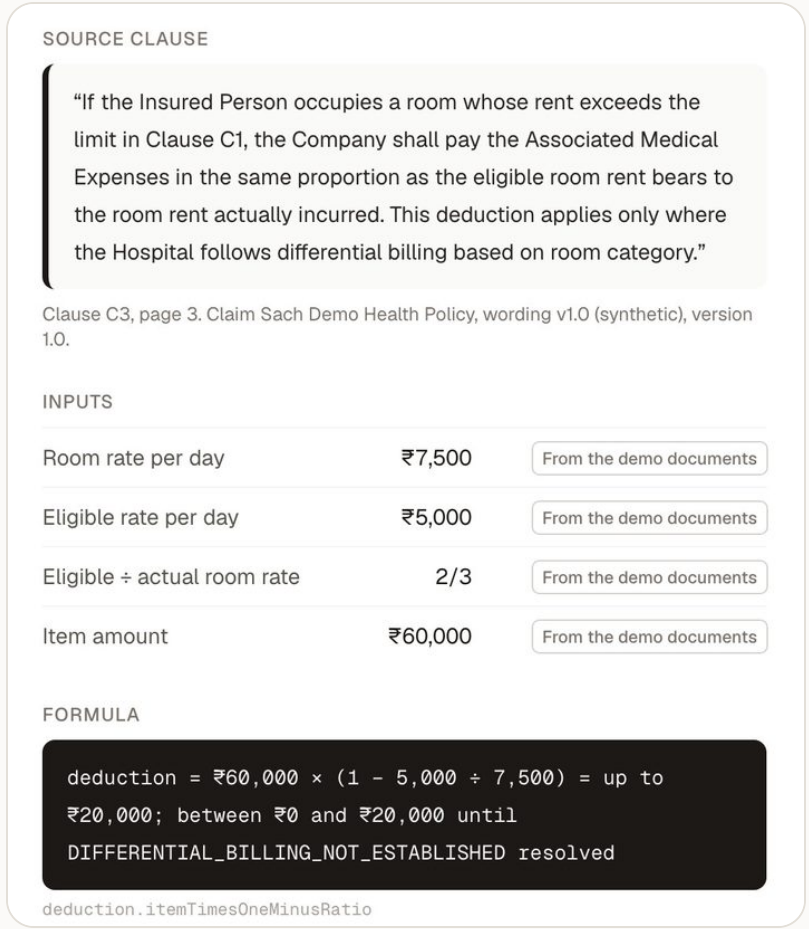
Unknown is never zero

A line the wording never classifies shows a dash and stays out of every total. A conditional amount shows both bounds and the condition. The policy terms this version does not model are listed on screen (appendix, slide 16).

The plain-words paragraph is generated from the trace by a deterministic explainer, not by a language model.



The drawer: result, state, and the explanation in plain words.



The clause, the inputs with provenance, the formula with the open code.

The same engine on a real wording, honestly

ASSOCIATED MEDICAL EXPENSES as defined by 4.1.1 Note (b)				
Nursing charges, 4 days	₹4,000	₹1,333 Calculated	₹1,333 Calculated	No change
Surgeon fees	₹60,000	₹20,000 Calculated	₹20,000 Calculated	No change
Anaesthetist fees	₹15,000	₹5,000 Calculated	₹5,000 Calculated	No change
Operation theatre charges	₹35,000	₹11,667 Calculated	₹11,667 Calculated	No change
Implant (knee prosthesis)	₹1,20,000	₹0 to ₹40,000 Conditional	₹0 to ₹40,000 Conditional	No change
Diagnostics	₹12,000	₹0 to ₹4,000 Conditional	₹0 to ₹4,000 Conditional	No change
Physiotherapy	₹6,000	₹2,000 Calculated	₹2,000 Calculated	No change

Arogya Sanjeevani, New India, UIN NIAHLIP20165V011920, from the regulator's copy: ten clauses verbatim, page-cited, SHA-256 on screen; partially encoded, synthetic schedule. Every figure is hand-checked on slide 22.

Capped percentage limits, 4.1(a) and 4.1(b)

Room 2% of the sum insured up to ₹5,000 a day; ICU 5% up to ₹10,000. For each category the applicable daily limit is the lower of the percentage-of-sum-insured amount and the stated daily cap; the trace shows both.

The ICU surprise, 4.1.1 Note (b)

A ₹15,000 ICU day exceeds its cap, so Note (b) applies the proportion even in the twin-sharing room: the surgeon's ₹20,000 shows in both columns. Differing ratios give bounds, not a number.

Co-payment as a bounded range, 9.5

5% of the admissible amount: ₹9,900 to ₹13,425 while implants and diagnostics are open; one figure only when everything else is settled.

Implants and diagnostics: a potential conflict with later norms, not a legal conclusion. IRDAI's circular IRDAI/HLT/REG/CIR/151/06/2020 of 11 June 2020 (clauses 4, 7 and 8) later kept implants, devices, diagnostics, pharmacy and consumables out of proportionate deduction for filings from 1 October 2020; the insurer's current sheet (UIN NIAHLIP25044V022425) keeps the same Note (b) sentence. Unverified either way, so the engine shows ₹0 to ₹40,000 and ₹0 to ₹4,000, "requires insurer confirmation", and the question sheet asks the insurer in writing. The full note, with the clauses quoted: appendix, slide 23.

Deterministic engine, AI only with human review

Frontend

TypeScript, strict. Next.js 16 (App Router), React 19, Tailwind 4. Six screens on a tested screen model; no business logic in components.

Rules engine

Pure TypeScript, no I/O, no model. Integer paise, exact ratios, one half-up rounding. A trace on every amount: rule, inputs, clause, formula, output, open assumptions. Six result states, eleven named uncertainty codes (appendix, slide 21).

Two wordings

A synthetic demo policy, and Arogya Sanjeevani (New India, from irdai.gov.in): ten clauses quoted verbatim, page-cited, SHA-256 shown. Partially encoded and labelled so.

AI, in its place

The Ruleset builder: a model proposes rules with quotes; the engine verifies each quote verbatim; a person confirms. Cached proposals in this build; the live adapter is off and was not exercised. Never in the calculation path.

Quality

Vitest, 181 tests in 18 files, written first. GitHub Actions on every pull request. Evaluation harness: six measures over two wordings, results committed and checked for staleness.

Runs anywhere

One command, offline on a laptop; fonts bundled; Vercel-ready. Nothing the user enters leaves the browser. No key, no account, no analytics, no database.

DATA FLOW, ALL IN THE BROWSER

```
fixture or upload (JSON, CSV) → ingestion: schema + verbatim evidence check → screen model → rules engine: classify → limits → proportion → co-payment → traces → six screens, room ladder, question sheet, evidence pack
```

AI seam, off by default: public clause text → model proposals → verbatim quote check → human confirmation → ruleset. No schedule, bill or name ever leaves the browser.

Item-level, clause-cited, honest about unknowns

1 Every number is clickable

Clause and page, inputs with where they came from, the formula, the operation. Tampered traces fail validation.

2 Unknown is never zero

Items the wording does not classify stay out of every total. A conditional amount shows both bounds and the condition.

3 Change your room, see the difference

Two rooms side by side, the rule behind every changed row; the ladder and the question sheet turn it into a decision and a call script.

4 Proven on a real wording, honestly

Arogya Sanjeevani, ten clauses verbatim with a checksum; where the 2019-20 text may conflict with IRDAI's 2020 norms, the engine flags it for insurer confirmation instead of deciding.

What we claim

Claim Sach turns a policy wording into a verifiable hospital-bill simulation: change the room, inspect every modelled deduction, and see what the documents cannot determine.

What we do not claim

Not the first to read clauses; not a payout guarantee; no accuracy figure beyond the measured ones. A bounded review of nine public sources found no tool doing this; we claim we looked, not that nobody does.

The room decision, made before admission

For the policyholder

The room decision with its deductions visible before admission, not after. Questions for the hospital and the insurer written out: what to ask, and what it is worth.

For a broker's claim-assistance agent

The clause on screen during the call. The uncertainty list is the call script; the evidence pack is the written answer.

Room rent	₹18,000	₹30,000	+₹12,000
Scenario total non-room items plus room rent	₹3,11,500	₹3,23,500	+₹12,000
Calculated deductions	**₹5,000**	**₹53,000**	+₹48,000
Conditional deductions	None	None	
Not classified shown separately, never totalled	₹7,500	₹7,500	
POTENTIAL DEDUCTIONS COVERED BY THIS SIMULATION	₹5,000 calculated, plus ₹7,500 not classified	**₹53,000 calculated, plus ₹7,500 not classified**	

One row decides it: ₹5,000 against ₹53,000 of calculated deductions once billing practice is confirmed.

MEASURED, NOT CLAIMED

MEASURE	SYNTHETIC	REAL WORDING
Clause retrieval, top-1	11 of 12	10 of 12
Rule extraction, field-level	10 of 10	10 of 12
Calculations, exact paise	9 of 9 · 46 amounts	3 of 3 · 29 amounts
Uncertainty and refusal	6 of 6	3 of 3
Proposal quotes verbatim (cached)	n/a	6 of 6
Proposals agree with reviewer (cached)	n/a	11 of 13

Measured on the synthetic wording and on ten encoded clauses of one real wording; proposals are cached, not a live run; results re-run on 25 October 2026, unchanged. No accuracy figure appears in the product. Every figure is a potential deduction covered by the simulation, never total out-of-pocket cost; every real-wording figure on these slides is hand-checked against the clauses on slide 22.

Where it fits Paytm, and what must be tested

VERIFIED FACTS

- Paytm Insurance Broking Pvt Ltd is a Direct (Life & General) broker: IRDAI Reg. No. 700, renewed 9 Feb 2026, valid to 16 Feb 2029.
- It distributes car, two-wheeler, health, life and term policies from 18+ insurers, including Niva Bupa and Aditya Birla Health. It does not underwrite.
- It already offers claim assistance by phone. Arogya Sanjeevani is a standardised product, but each ruleset is verified against the exact insurer, UIN, policy wording and version before use; one reviewed encoding is a head start, never a licence.
- FY25 turnover ₹37.1 crore, loss ₹13.2 crore (Form AOC-1), inside a ₹2,593 crore financial-services line in FY26. Insurance is a small line for Paytm today; clarity is a wedge worth testing.

HYPOTHESES, EACH WITH ITS TEST

HYPOTHESIS	TEST
Fewer claim-time support contacts and disputes	Four-week pilot at one claim-assistance desk, against a matched cohort
More eligible-room choices when the consequence is visible	Room-change rate in the simulator, then claims data
Panel insurers co-fund a coverage-clarity tool	Three conversations; one signed pilot letter is the pass
A coverage check lifts conversion or renewal on the policy page	A/B test, if Paytm chooses to run one

Revenue option · untested

Broker-side feature

Inside the health-policy journey and claim assistance; the return is retention and fewer disputes.

Revenue option · untested

Insurer-paid integration

Panel insurers co-fund coverage clarity for the policies they sell; a signed pilot letter is the pass.

Revenue option · untested

Premium claim-assistance tier

The question sheet and the evidence pack as part of a paid assistance service.

Which option, if any, depends on the tests above. No Paytm integration is claimed or built. Bima Sugam is not live; it belongs in the future tense.

Built, tested, runs with the Wi-Fi off. Here is the line between done and proposed.

BUILT AND VERIFIED TODAY

- Six screens, Coverage Twin, room ladder, question sheet, evidence pack; one command, offline.
- A deterministic rules engine with a validated trace on every amount; unknown is never zero.
- Two wordings: a synthetic policy and ten clauses of Arogya Sanjeevani (New India), quoted verbatim with a checksum.
- 181 tests, CI on every pull request, an evaluation harness with committed results, a hand audit of every real-wording figure over jointly possible outcomes.
- The ruleset builder with cached model proposals, each quote verified verbatim, each one confirmed by a person.

PROPOSED, NOT BUILT

- A live model adapter: the route exists, is off by default, returns 503 unless configured, and was never exercised.
- A second real wording taken through the builder with that adapter on.
- Independent review of the evaluation fixtures and of the clause readings.
- A four-week pilot at a broker's claim-assistance desk; any Paytm integration at all.

Shortlist us and we bring a laptop. Three minutes: change the room, open a number, watch the engine refuse to guess on a real wording.

Code, tests, documents and the calculation audit: github.com/Pushkarmondal/paytm-hackathon · the proof, problem by problem, follows as an appendix.

Pushkar Mondal · Founding Engineer, Xeliport · October 2026

The proof, problem by problem

Slides 1 to 11 are the pitch. These show the working product on each problem, with whole-element screenshots from the running app. Every entry is a link.

13	**The eleven problems, mapped**	What goes wrong, what the product does, where
14	**Which items are cut** problem 2	Classification by the wording's own definition
15	**A hospital fact** problem 3	Differential billing: a range, then a number recorded as yours
16	**Unknown is never zero** problem 4	The six result states and the not-modelled list
17	**The ruleset builder** problem 9	A model proposes, the engine checks the quote, a person confirms
18	**Your own documents** problem 10	Upload, provenance on every field, the verification record
19	**Decide the room, then ask** problem 11	Room ladder and question sheet
20	**Why it changed; the evidence pack**	Per-rule diffs and verbatim lines from the export
21	**Engineering guarantees**	The engine's result and uncertainty types, verbatim
22	**Calculation audit**	Every real-wording figure derived by hand from the clauses
23	**The regulatory note in full**	The 2020 circular quoted, the current UIN, what is and is not verified

Eleven problems, and what the product does about each

#	WHAT GOES WRONG FOR THE POLICYHOLDER	WHAT CLAIM SACH DOES, AS BUILT	WHERE · SLIDE
01	A room above the eligible rate cuts the surgeon's fee, not just the room	Applies the proportion (eligible ÷ actual) item by item, with the ratio and the clause on each row	Coverage Twin · 4
02	Which items are cut depends on this wording's own definition	Every line classified by the policy's text: included, excluded, or unknown; never one ratio over the whole bill	Twin · 14
03	A fact only the hospital knows: does it bill differently by room?	A range with both bounds until it is confirmed; the answer is recorded as yours in every trace	Uncertain, Input review · 15
04	Items the wording never mentions get quietly treated as ₹0	Unknown is never zero: shown apart from every total, with who can settle it; the not-modelled list is on screen	Uncertain · 16
05	ICU rates and capped percentage limits behave differently from room rent	Lower of percent and cap; ICU can trigger the proportion; two different ratios give bounds, not a guess	Real wording · 6
06	A co-payment stacks on everything else	A bounded range until every other amount is settled; a single figure only then	Real wording · 6
07	An older wording may conflict with later IRDAI norms	A caveat on the affected items: requires insurer confirmation, with the circular and clause named; a potential conflict, not a legal finding	Real wording · 6
08	“How did you get this number?”	A trace on every amount: clause and page, inputs with provenance, formula, result; tampered traces fail validation	Trace drawer · 5
09	Encoding a new wording takes an expert	Find the clause by keyword; a model proposes rules with quotes, the engine verifies each quote verbatim, a person confirms	Ruleset builder · 17
10	Your real documents, not a demo, and no server to trust	JSON and CSV read in the browser only; schema-checked; misquotes refused; every field correctable with provenance	Input review · 18
11	Which room, what to ask, and how to prove it later	Room ladder, question sheet per hospital and insurer, why-this-changed, and an evidence pack with every trace	Ladder, sheet, pack · 19, 20

Which items are cut is the wording's call, not ours

PROBLEM 2 THE POLICY'S OWN DEFINITION DECIDES

NOT SUBJECT TO THE PROPORTIONATE DEDUCTION				
Implant (knee prosthesis)	₹1,20,000	₹0 Not subject to deduction	₹0 Not subject to deduction	No change
Pharmacy and consumables	₹25,000	₹0 Not subject to deduction	₹0 Not subject to deduction	No change
Diagnostics	₹12,000	₹0 Not subject to deduction	₹0 Not subject to deduction	No change
NOT CLASSIFIED BY THE WORDING				
Physiotherapy	₹6,000	— Cannot determine	— Cannot determine	—
Hospital admin charges	₹1,500	— Cannot determine	— Cannot determine	—

The lower half of the same Coverage Twin table: three lines the demo wording keeps whole, two it never classifies. Rows that cannot be determined show a dash, never ₹0.

Included: by the wording's definition

The synthetic clause C4 names nursing, surgeon, anaesthetist, consultant and operation-theatre charges. Arogya Sanjeevani's Note (b) says "all other expenses incurred at the Hospital, with the exception of cost of medicines". Same engine, two different lists.

Excluded: by the same wording

Implant, pharmacy and diagnostics stay whole under C4. Under the real wording only medicines are excepted, so implants and diagnostics are reduced, with a caveat (slide 6). No universal Indian policy is hard-coded anywhere.

Unknown: out of every total

Physiotherapy and hospital admin charges are neither named nor excluded by C4. They are shown separately with "Cannot determine". On Input review you can classify them yourself; the choice is recorded as your assumption in every trace.

A hospital fact turns a range into a number

Ask the hospital

A fact about billing practice that only the hospital can state.

The estimate does not state whether the hospital bills differently by room category. C3 applies only where it does. Until this is established, the proportionate deduction is between ₹0 and ₹38,000.

Affects Nursing charges, 4 days, Surgeon fees, Anaesthetist fees, Operation theatre charges

At stake ₹0 to ₹38,000 · Not established whether the hospital follows differential billing by room category: low = no deduction (C3 does not apply), high = the full proportionate deduction

Resolves it Ask the hospital's billing desk whether the room category changes the rates of these items, then record the answer on Input review.

`DIFFERENTIAL_BILLING_NOT_ESTABLISHED`

Open question, four items affected, ₹0 to ₹38,000 at stake, and what to ask.

ASSOCIATED MEDICAL EXPENSES as defined by C4

Nursing charges, 4 days	₹4,000	₹0 Calculated	₹1,333 Calculated	+₹1,333
Surgeon fees	₹60,000	₹0 Calculated	₹20,000 Calculated	+₹20,000
Anaesthetist fees	₹15,000	₹0 Calculated	₹5,000 Calculated	+₹5,000
Operation theatre charges	₹35,000	₹0 Calculated	₹11,667 Calculated	+₹11,667

After "Yes, confirmed by me" on Input review: the same four rows, now calculated.

PROBLEM 3 DIFFERENTIAL BILLING BY ROOM CATEGORY

Before: both bounds, each with its own trace

Clause C3 applies only where the hospital bills by room category; the estimate does not say. The surgeon's fee shows ₹0 to ₹20,000: low = C3 does not apply, high = the full proportion. The headline keeps it out of the calculated figure.

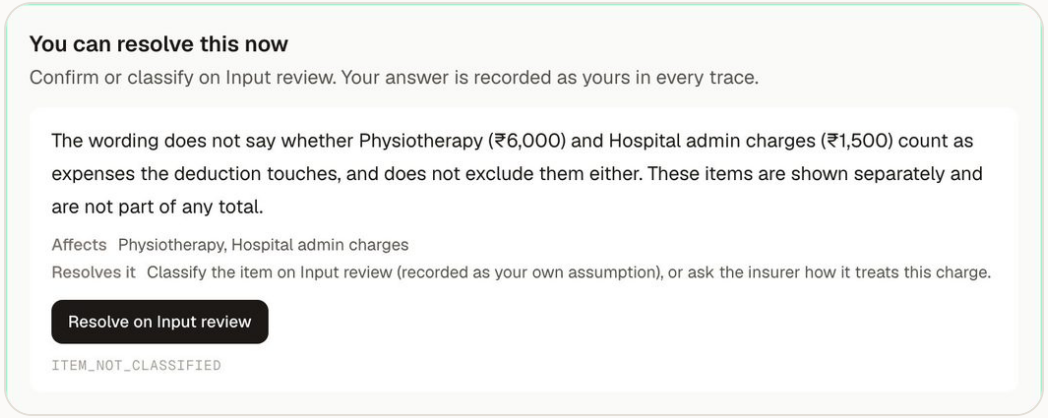
After: one answer, recorded as yours

Confirm it on Input review and every dependent amount recomputes: calculated deductions move from ₹15,000 to ₹53,000, the open questions drop from 3 to 2, and every trace shows the input as "confirmed by you".

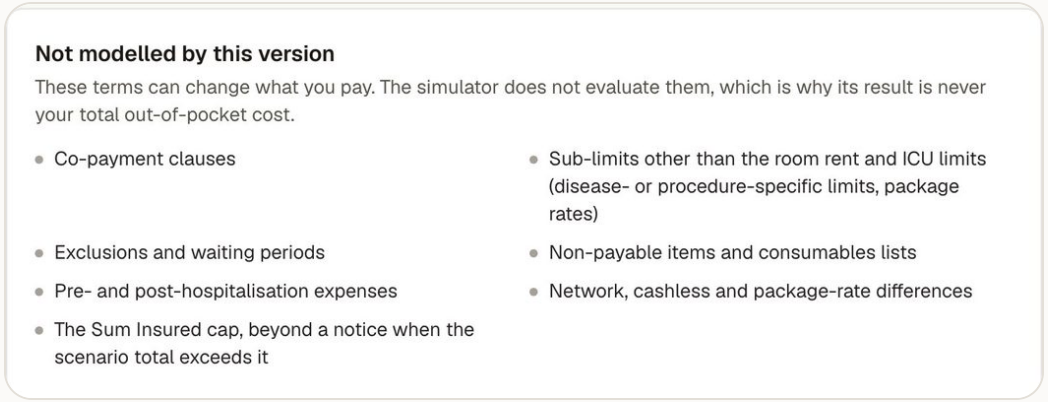
Ranges only with defensible bounds

A yes-or-no fact has two computable outcomes, so both are shown. An item the wording never classifies has none, so it stays unresolved and out of every total.

Unknown is never zero



Two lines the wording never classifies, shown apart with the reason and the fix.



Every term this version does not evaluate, where the user cannot miss it.

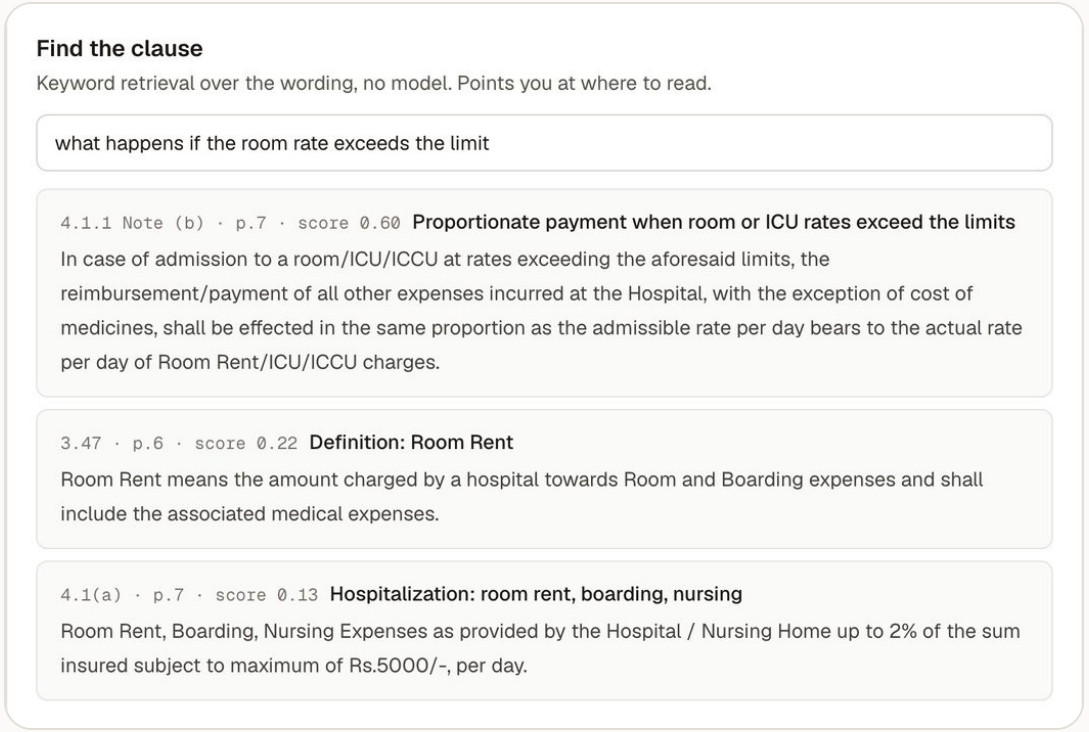
PROBLEM 4 SILENCE IN THE WORDING IS NOT ₹0

STATE	MEANING, AND HOW IT IS SHOWN
Calculated	Every input established or confirmed by you, all conditions met. Plain ink; click for the trace.
Conditional	The rule applies if a named condition holds. Amber badge; a range with both bounds.
Cannot determine	The documents lack or contradict an input the rule needs. Grey badge; a dash, never 0.
Not subject to deduction	The policy's own definition keeps the item out of this rule. Neutral badge; the item stays in the bill.
Requires insurer confirmation	The policy is silent, or only the insurer can settle it. Blue badge; listed with what to ask.
Review required	The schedule contradicts the wording. No amount until a person looks.

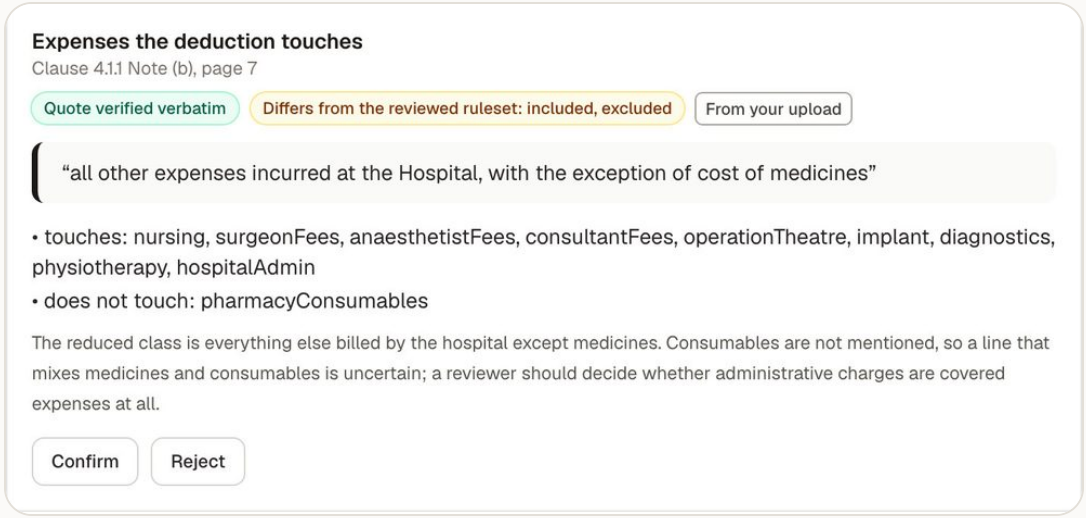
In the code, Money has no unknown value: an unknown amount is the absence of a Money plus an uncertainty code. A total with an unknown inside reads "₹15,000 calculated, plus ₹7,500 not classified", never one figure.

Model proposes, engine checks, you confirm

PROBLEM 9 ENCODING A NEW WORDING



Find the clause: keyword retrieval over the wording, no model, scored and page-cited. It points a reviewer at where to read.



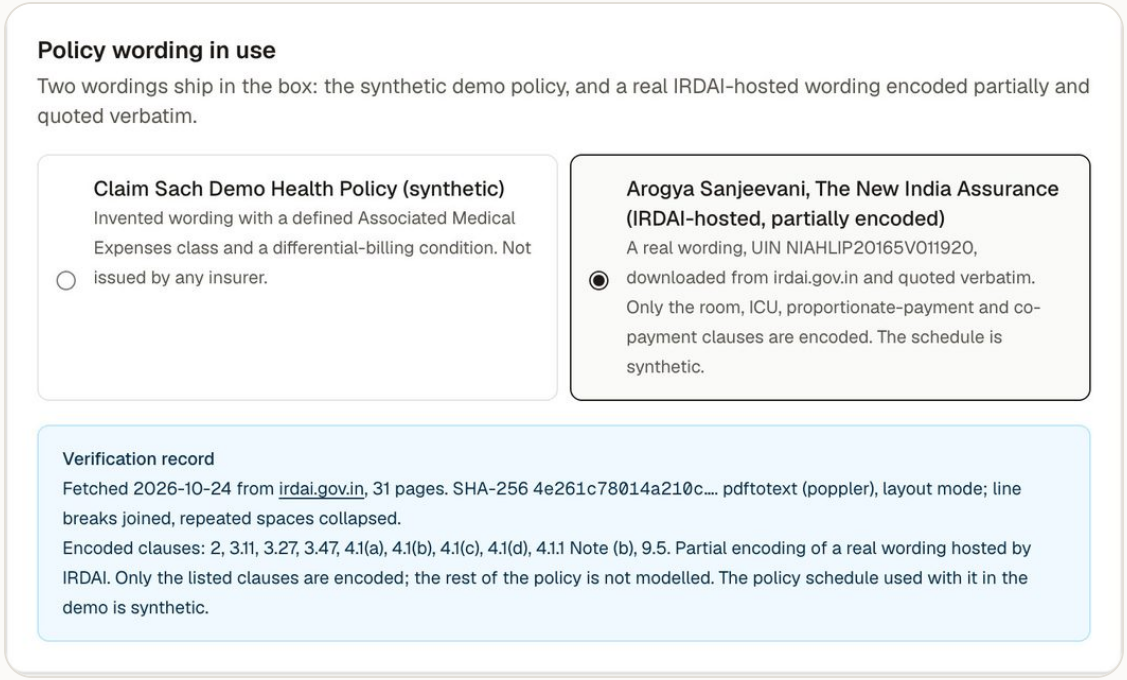
One cached proposal for Arogya Sanjeevani. Three checks on every card: the quote verified verbatim against the clause; agreement with the human-reviewed ruleset, and here it differs; and its provenance.

The boundary

Confirmed, verified proposals become the active ruleset with the full clause as evidence; a misquote cannot be confirmed. Proposals here are cached and say so; the live adapter returns 503 unless configured and was not exercised. The model never calculates and never sees a schedule, a bill or a name.

Your own documents, and where every value came from

PROBLEM 10 REAL DOCUMENTS, NO SERVER TO TRUST



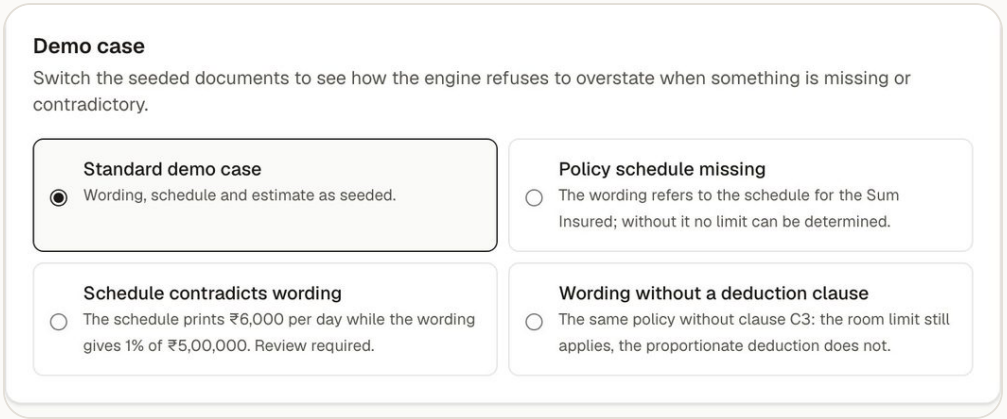
Input review with the real wording selected: the verification record carries the fetch date, source, page count, SHA-256, extraction method and the list of encoded clauses.

Upload, in the browser only

A ruleset (JSON), a schedule (JSON) and a hospital estimate (JSON or CSV), up to 5 MB each, read with the File API and never sent anywhere. Schema-checked; a rule that misquotes its clause is refused.

Provenance on every field

Every value carries a badge: demo documents, your upload, confirmed by you, or entered by you. Correct one and every trace updates. No account, key, analytics or storage.

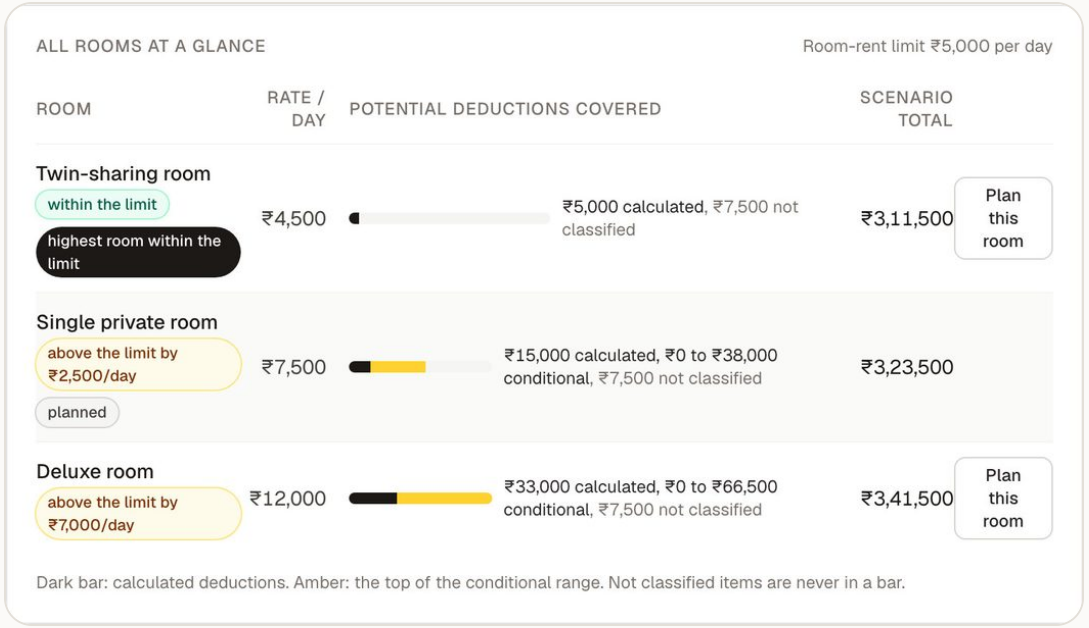


Three failure cases ship as demo variants: the engine refuses to overstate.

Decide the room, then know what to ask

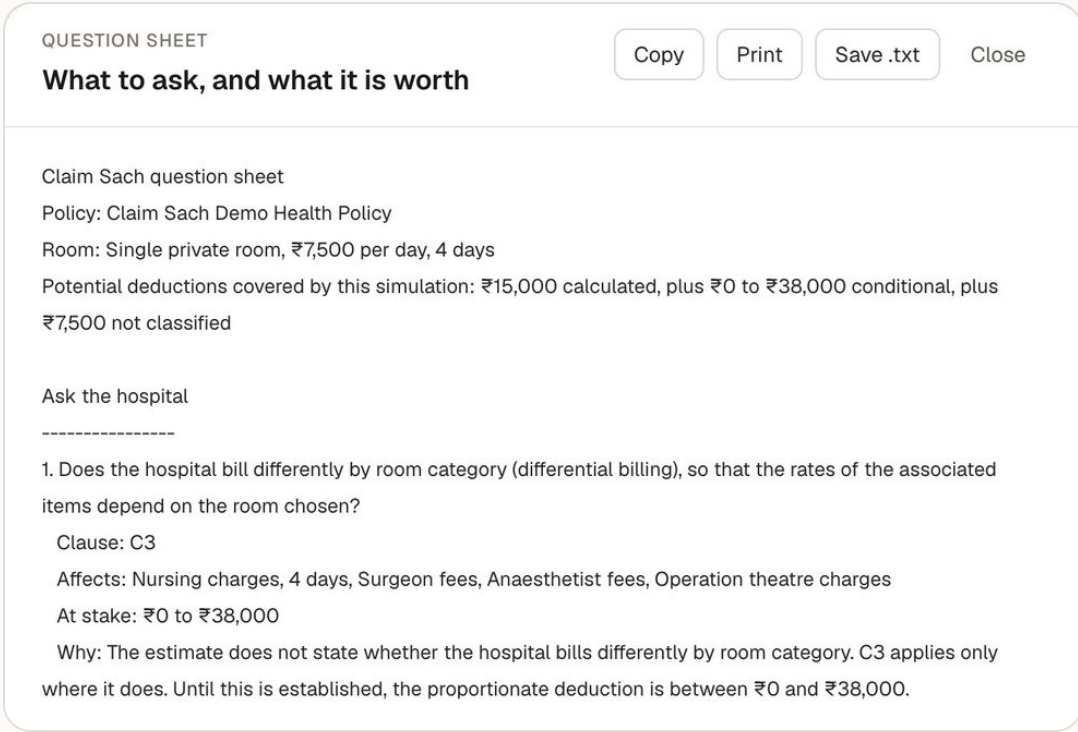
Room ladder

Every room on one view with the deductions at each; the highest room within the limit is marked. One click plans it. The product shows; it does not recommend.



Question sheet

Every open uncertainty becomes a question for the hospital or the insurer, with its clause and the rupees at stake. Copy, print or save; take it to the billing desk.



PROBLEM 11 WHICH ROOM, AND WHAT TO ASK

Why it changed, and the evidence to take with you

ROOM RENT	SCENARIO TOTAL	CALCULATED DEDUCTIONS	CONDITIONAL DEDUCTIONS
+₹12,000	+₹12,000	**+₹10,000**	None → ₹0 to ₹38,000

RULE

Room rent limit: amount above the limit `C1.roomRentLimit`

SOURCE CLAUSE

> "Room rent and boarding charges incurred during Hospitalisation are payable up to 1% of the Sum Insured per day, as stated in the Policy Schedule."

Clause C1, page 2. Claim Sach Demo Health Policy, wording v1.0 (synthetic).

WHAT DIFFERS BETWEEN A AND B

Input	A	B
Room rate per day	₹4,500	₹7,500

`A: excess = max(0, ₹4,500 − ₹5,000) × 4 days = ₹0`

`B: excess = max(0, ₹7,500 − ₹5,000) × 4 days = ₹10,000`

AFFECTED ROWS

Item	A	B	Change
Room rent above the limit, 4 days	₹0 Calculated	₹10,000 Calculated	+₹10,000

Per rule: the clause, the inputs that differ, both formulas, the affected rows.

```
# Claim Sach evidence pack
## Scenario
- Room: Single private room, ₹7,500 per day, 4 days (room rent ₹30,000)
## Potential deductions covered by this simulation
**₹15,000 calculated, plus ₹0 to ₹38,000 conditional, plus ₹7,500 not classified**
(overall state: conditional). Not the total out-of-pocket cost.
## Items
| Item | Amount | Classification | Deduction | State |
| Surgeon fees | ₹60,000 | includedInDeduction | ₹0 to ₹20,000 | Conditional |
| Physiotherapy | ₹6,000 | unknown | — | Cannot determine |
… 8 rows omitted …
## Traces
### Room rent limit (C1.roomRentLimit)
> Room rent and boarding charges incurred during Hospitalisation are payable up to
1% of the Sum Insured per day, as stated in the Policy Schedule.
Formula: `eligiblePerDay = 1% × ₹5,00,000 = ₹5,000 per day`
Result: ₹5,000 (Calculated)
… every other trace follows, then the verification record and the disclaimer …
```

Verbatim lines from the exported Markdown evidence pack (JSON is one click away); "…" marks omitted lines.

Six states, eleven named uncertainties, no silent zeros

```ts
export type ResultState =
  | "calculated"
  | "conditional"
  | "unknown"
  | "excluded"
  | "requiresInsurerConfirmation"
  | "reviewRequired";

export type UncertaintyCode =
  | "SCHEDULE_MISSING"
  | "SCHEDULE_CONTRADICTS_WORDING"
  | "DIFFERENTIAL_BILLING_NOT_ESTABLISHED"
  | "ITEM_NOT_CLASSIFIED"
  | "WORDING_SILENT_ON_ROOM_UNAVAILABILITY"
  | "RULE_ABSENT"
  | "LIMIT_RULE_ABSENT"
  | "ICU_RATE_NOT_STATED"
  | "RATIO_AMBIGUOUS_ROOM_AND_ICU"
  | "WORDING_PREDATES_2020_NORMS"
  | "POLICY_CAVEAT";

// src/engine/types.ts, verbatim. Every uncertainty the
// engine can raise is a named code; every amount has a state.
```

Integer paise

Money is a branded integer of paise. mulRatio multiplies with BigInt and rounds half up once per operation; nothing is a float.

Exact ratios

Eligible ÷ actual stays a reduced fraction (gcd), so 5,000 ÷ 7,500 is 2/3, not 0.6667.

No universal policy

Each wording is its own ruleset with its own definitions. No rule is hard-coded; no single ratio is ever applied to a whole bill.

Unknown has no Money

An unknown amount is the absence of a Money plus an uncertainty code. It cannot be added, so it cannot be totalled by accident.

Traces are validated

validateTrace checks each trace against the verbatim clauses and schedules; a tampered or misquoted trace fails. Uploaded rulesets get the same check.

Tests first

181 Vitest tests in 18 files, written before the code. CI runs lint, typecheck, tests, build and the evaluation harness on every pull request, and fails if the committed results are stale.

Every real-wording figure, derived by hand from the clauses

STEP	CLAUSE	ARITHMETIC	RESULT
The bill: ten non-room lines		ICU 15,000 + nursing 4,000 + surgeon 60,000 + anaesthetist 15,000 + OT 35,000 + implant 1,20,000 + pharmacy 25,000 + diagnostics 12,000 + physio 6,000 + admin 1,500	**₹2,93,500 · + room ₹30,000 = ₹3,23,500**
Eligible room rate	4.1(a)	min(2% × ₹5,00,000 = ₹10,000, ₹5,000)	**₹5,000 a day**
Room excess, single private	4.1(a)	(₹7,500 − ₹5,000) × 4 days	**₹10,000**
Eligible ICU rate	4.1(b)	min(5% × ₹5,00,000 = ₹25,000, ₹10,000)	**₹10,000 a day**
ICU, 1 day at ₹15,000: excess	4.1(b)	(₹15,000 − ₹10,000) × 1 day; the ICU line is governed by its own limit and is never in the Note (b) class	**₹5,000**
Ratio	Note (b)	room 5,000 ÷ 7,500 = 2/3; ICU 10,000 ÷ 15,000 = 2/3; same, applies	**deduction = item ÷ 3**
Nursing · surgeon · anaesthetist · OT · physio	Note (b)	4,000 ÷ 3 · 60,000 ÷ 3 · 15,000 ÷ 3 · 35,000 ÷ 3 · 6,000 ÷ 3	**1,333.33 · 20,000 · 5,000 · 11,666.67 · 2,000**
Implant · diagnostics (caveat)	Note (b)	₹0 to 1,20,000 ÷ 3 · ₹0 to 12,000 ÷ 3	**0 to 40,000 · 0 to 4,000**
Pharmacy and consumables · hospital admin	Note (b)	medicines excepted, consumables not addressed · covered expense not stated	**not classified, ₹26,500 apart**
Calculated deductions		10,000 + 5,000 + 1,333.33 + 20,000 + 5,000 + 11,666.67 + 2,000	**₹55,000.00**
Co-payment, low and high	9.5	5% × (3,23,500 − 55,000 − 44,000 − 26,500) · 5% × (3,23,500 − 55,000)	**₹9,900 · ₹13,425**
Conditional deductions, jointly		low: no reduction, unclassified not admissible: 0 + 5% × 2,42,000 · high: reduction applied, unclassified admissible: 44,000 + 5% × 2,24,500	**₹12,100 to ₹55,225**
Twin-sharing room		room excess 0; ICU triggers the same 2/3: 5,000 + 13,000 + 20,000 + 5,000 + 2,000 · jointly 0 + 5% × 2,40,000 to 44,000 + 5% × 2,22,500	**₹45,000 · ₹12,000 to ₹55,125**
Deluxe room, ₹12,000		ratios 5/12 and 2/3 differ: × 1/3 to × 7/12; 28,000 + 5,000 calculated · jointly 40,000 + 5% × 2,42,000 to 1,47,000 + 5% × 1,61,500	**₹33,000 · ₹52,100 to ₹1,55,075**

Sum insured ₹5,00,000 (synthetic schedule); four room days. Conditional bounds are taken over every jointly possible outcome (sixteen for the deluxe room), so no bound pairs a maximum deduction with a co-payment computed as if it had not happened. Integer paise, reduced fractions, one half-up rounding; the display drops paise. The engine run on 25 October 2026 matches; the harness checks 29 amounts. Full working: docs/deck/CALCULATION-AUDIT.md.

The circular, the two wordings, and what is verified

IRDAI, "Norms on Proportionate Deductions", Ref IRDAI/HLT/REG/CIR/151/06/2020, 11 June 2020

Read from irdai.gov.in on 24 October 2026; text extracted from a scanned PDF, OCR artefacts removed.

Clause 4. "The following expenses are not allowed to be part of the definition of 'associate medical expenses'. a. Cost of pharmacy and consumables; b. Cost of implants and medical devices c. Cost of diagnostics"

Clause 6. "Insurers shall ensure that proportionate deductions are not applied in respect of the hospitals which do not follow differential billing or for those expenses in respect of which differential billing is not adopted based on the room category."

Clause 7. "Insurers are not permitted to apply proportionate deduction for 'ICU charges' as different categories of ICU are not there."

Clause 8. Products filed "on or after 01st October, 2020"; existing contracts "modified as and when they are due for renewal from 01st April, 2021 onwards."

Superseded

The Master Circular on Health Insurance Business, Ref IRDAI/HLT/CIR/PRO/84/5/2024, 29 May 2024, supersedes the circulars in its Annexure-6; item 19 is this one. Whether the norm continues under the 2024 regime is not verified here.

The 2019-20 wording this demo encodes

UIN NIAHLIP20165V011920, page 7, Note (b): "all other expenses incurred at the Hospital, with the exception of cost of medicines, shall be effected in the same proportion as the admissible rate per day bears to the actual rate per day of Room Rent/ICU/ICCU charges." On its face that reaches implants and diagnostics.

The insurer's current sheet

"Customer Information Sheet AROGYA SANJEEVANI POLICY, master w.e.f. 01 10 2024", newindia.co.in, read 25 October 2026: UIN NIAHLIP25044V022425 on every page, the same Note (b) sentence, and a note that the policy document prevails. The full current wording was not fetched, encoded or verified.

What the engine does with this

Implants and diagnostics carry the caveat WORDING_PREDATES_2020_NORMS: ₹0 (reduction not applied) to the full proportion, "requires insurer confirmation", and a written question on the question sheet; totals bounded over the joint outcomes (slide 22).

Not verified, and not claimed

How the insurer settles this item under either version; whether a given contract was modified at renewal; any other insurer's wording. Each ruleset is verified against the exact insurer, UIN, wording and version before use.